



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

January 23, 2026

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

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Finite Element Method

Practical Implementation

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Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

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Outlook

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$$\ell \gg \lambda$$

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$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

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 - ▶ $R > 4000$: Kolmogorov Energy Cascade (fully developed phase)

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- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

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- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla) u)$

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Lid Driven Cavity Flow

Pressure Driven Channel Flow

Nonuniform Periodic Flow

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

- Fundamental Definitions
- Basic Equations
- Examples

Numerical Approaches

- Finite Difference Method
- Finite Volume Method
- Finite Element Method

Practical Implementation

- Kelvin-Helmholtz Instability
- Rayleigh-Taylor Instability
- Outlook

Conceptual Overview

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 - mass and energy can leak due to numerical errors

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

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Conceptual Overview

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

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Outlook

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Outlook

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Outlook

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Thank You!

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Resources on GitHub: [here](#)